

Measuring and Modeling Stage Placement in Hybrid Retrieval-Augmented Generation across Edge, LAN, and WAN

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1 Background and Current State

We have built and deployed a working *hybrid* Retrieval-Augmented Generation (RAG) system: sparse keyword retrieval (BM25 via Tantivy), dense semantic retrieval (BGE-M3 embeddings searched in Qdrant), reciprocal-rank fusion, and LLM generation (Llama-3-8B-Instruct, AWQ 4-bit, served by vLLM [14, 13]). The corpus is the full MS MARCO passage collection [12]: **8,841,823 passages** embedded at 1024 dimensions, HNSW-indexed in Qdrant, ingested at roughly 538 passages per second on an RTX 3090. The system is deployed on two LAN machines connected over gRPC (WireGuard): a gateway node (embedding, dense+sparse retrieval, fusion) on an Intel i7-10750H laptop with GTX 1660 Ti, and a generation node (document lookup, vLLM) on an RTX 3090 host. An earlier three-node version placed a third, low-power laptop (Intel CPU, 16 GB RAM) at the network edge as entry point and sparse-retrieval engine, across a city-wide link treated as the WAN leg.

2 A Failed Hypothesis That Points at a Real Gap

The original design hypothesized a 30–40% TTFT (time-to-first-token) reduction by running edge-side sparse retrieval in parallel with transit of the query toward the other nodes: $T_{\text{retrieval}} = \max(T_{\text{sparse}}, T_{\text{WAN}} + T_{\text{dense}})$. Our instrumented three-node runs showed nothing like it. The surviving 100-query log records sparse retrieval at 0.6–10 ms against TTFT of 51–227 ms: the stage we chose to distribute was one to two orders of magnitude cheaper than the pipeline it was meant to accelerate. Since parallel overlap can save at most $\min(T_{\text{sparse}}, T_{\text{WAN}} + T_{\text{dense}})$, no link latency could have delivered the target — a short WAN leg offered little to hide, and a long one would only have slowed the dense branch. One caveat: we can no longer reconstruct where the machines physically were during those logged runs, so we treat the 5–15 ms round-trip from our mid-evaluation report as descriptive, not tied to this log; the new campaign will record placement and network metadata with every measurement. The three-node setup itself ended for operational reasons, not hardware: the edge laptop belonged to a groupmate who was often unavailable, so we consolidated its duties into the gateway and the two-node system worked fine.

That failure motivates this proposal: placement by intuition silently costs performance, the stages worth distributing are the *expensive* ones (dense retrieval, document handling, generation — not sparse lookup), and topology change is a normal event in real deployments. Splitting is often not a choice: large models need GPUs most machines do not have, data-residency rules can force retrieval and generation onto different hosts, and cost pushes work toward whatever hardware exists. Yet the literature, as far as we can find, cannot answer: *where should each stage of a hybrid RAG pipeline run, under which network conditions, and what exactly do we lose when the topology degrades?*

3 The Gap in the Literature

DNN partitioning between device and cloud is well studied at the *layer* level (Neurosurgeon [1], PipeEdge [2]); LLM serving splits model *phases* across machines (Splitwise [3]). For RAG specifically, existing work optimizes KV caching of retrieved knowledge (RAGCache [4], TurboRAG [5]), serving-system optimization (RAGO [6], METIS [7]), index management on a single host (EdgeRAG [8], VectorLiteRAG [9]), or data distribution across edge/cloud tiers (EACO-RAG [10], DGRAG [11]). We could not find a measurement study and placement policy for *which pipeline stage runs on which machine* in a multi-node hybrid RAG deployment: the five stages (sparse, dense, fusion, hydration, generation) can each sit on a different node, and the right answer changes with RTT, bandwidth, and node capability. Node loss (what happens to latency and answer quality when a node drops out and roles consolidate) appears entirely unmeasured for RAG.

4 Research Questions

- **RQ1 (Placement).** Which placement for each stage minimizes TTFT and end-to-end latency under a given network regime (RTT, bandwidth), and where does the best placement change?
- **RQ2 (Degradation).** When a node drops out — hardware failure or simple unavailability (our real three-to-two consolidation) — what is the cost in latency and retrieval quality, and how fast can the system adapt?
- **RQ3 (Policy).** Can a simple analytical cost model — per-stage compute times, payload sizes, network terms — predict the best placement within a stated tolerance (e.g. TTFT within 15% of measured), and can the running system use it to pick placement per network regime without the full experiment matrix?

5 Method

Testbed. The existing heterogeneous nodes (RTX 3090 host, GTX 1660 Ti laptop, revived low-power laptop) over WireGuard; the full 8.84M-passage index is already in place. **Instrumentation.** We will extend the existing per-stage timing logs into a structured benchmark harness (per-stage milliseconds, TTFT, decode throughput, true stream end-to-end) and add retrieval-quality evaluation against MS MARCO qrels (recall@ k) plus an answer-level check (exact match), so topology-induced damage shows up in generated answers and not only in retrieval; the sparse index will be scaled from a small test index to the full corpus. **Network control.** Today the only delay knob is an application-level header; we will add kernel-level per-link shaping (Linux netem on each WireGuard interface: delay and bandwidth, 0/10/40/80 ms RTT) so regimes shape every hop, and re-validate key configurations over the genuine WAN link. **Placement matrix.** Stage-to-node assignment becomes a configuration switch (fusion on gateway vs. generator; hydration local vs. remote; sparse retrieval at the edge vs. LAN), reconstructing the archived three-node code paths. **Campaign.** A factorial sweep over placement $\times$ network regime $\times$ retrieval depth with repeats and percentile statistics; a degradation experiment kills nodes mid-load to measure failover cost; the RQ3 cost model is fit on half the configurations, validated on the held-out half, and then run live, letting the model pick placement per regime against the fixed ones. **Limitations.** Three machines, one per capability tier: results characterize this class of testbed, not a fleet; netem regimes bound generality.

6 Expected Contributions

(1) To our knowledge, the first factorial measurement of stage-to-node placement for hybrid RAG on commodity heterogeneous hardware under controlled WAN regimes; (2) a measurement of what topology degradation actually costs, grounded in our own three-to-two-node consolidation; (3) a cost-model placement policy, validated on held-out configurations and exercised live; (4) an open, reproducible testbed with all artifacts.

7 Timeline and Target Venues

Weeks 1–2: harness, full-corpus sparse index, quality evaluation. Week 3: placement matrix and network emulation. Weeks 4–5: campaigns, cost model, analysis. Weeks 6–8: writing, internal review, submission. If time runs short we cut in order: live policy validation, extra network regimes — never the degradation study or the quality metrics. We target Q1 journals with rolling submission: *Future Generation Computer Systems*, *Computer Networks*, or *IEEE Transactions on Cloud Computing* (all currently Q1 in SCImago/JCR rankings). A conference extension (e.g. IEEE ICDCS 2027, call expected around December 2026) remains possible for the policy contribution.

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